Yirou Dong

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Education

University of Edinburgh – GPA: 89% (first year) Edinburgh, UK
Artificial Intelligence BSc *Sep. 2024 – May 2028*
Relevant courses in progress: Introduction to Algorithms and Data Structures, Foundations of Data Science, Reasoning and Agent, Software Engineering and Professional Practice

Relevant courses completed: Discrete Mathematics and Probability(76%), Introduction to Computation(100%), Fundamentals of Algebra and Calculus(100%), Introduction to Linear Algebra(98%), Cognitive Science(73%), Object Oriented Programming(92%), Calculus and its Applications(73%)

Experience

Planning and Control Team Member Oct 2025 – Present
Edinburgh University Formula Student (EUFS) *Edinburgh, UK*

- Rapidly onboarded onto the team's ROS2-based software stack and development tools.
- Contributed to codebase maintenance and optimization projects.
- Developed automation scripts (e.g., for build/testing) that reduced manual setup time, accelerating team iteration cycles.

Projects

Hybrid Chess Analyzer | *Python, Google Gemini API, Gradio, Stockfish* Dec. 2025 – Feb. 2026

- Integrated Google Gemini 1.5 Pro to build a chess coaching system with natural language understanding.
- Designed JSON-based function calling to enable reliable tool use despite Gemini API limitations.
- Implemented session-based state tracking for board position persistence across conversations.
- Engineered multi-move parsing from single user inputs, demonstrating fine-grained NLP.
- Built modular architecture with Gradio UI and Stockfish 16 for real-time analysis.

Data Visualization Critique & Redesign | *Python, Pandas, Matplotlib, Seaborn* Oct. 2025 – Nov. 2025

- Critiqued an ineffective data visualization against established principles, identifying 7+ flaws in data-ink ratio, labeling, and clarity.
- Engineered a new, clear multi-plot visualization to accurately compare city metrics (density, universities, bike commute%), enabling immediate cross-city analysis.
- Derived and summarized key insights, such as Edinburgh's leading bike commute rate, demonstrating the ability to translate data into narrative.

EVE Transport Cost Estimation Engine | *Java* Oct. 2025 – Nov. 2025

- Designed and implemented a robust, object-oriented data processing engine to calculate logistics costs across land, sea, and air transport modes.
- Architected the system with a focus on data integrity, implementing comprehensive input validation and error-handling logic, ensuring system robustness and data integrity under various test scenarios.
- Demonstrated the ability to model complex business rules into a well-structured and maintainable application.

Handwritten Digit Recognition System | *Python, TensorFlow, OpenCV, Numpy, Matplotlib* Jul. 2025

- A deep learning-based handwritten digit recognition system implemented with TensorFlow and Keras, trained on the MNIST dataset.
- Tech Stack: Built a Convolutional Neural Network model using Python and TensorFlow/Keras.
- End-to-End Pipeline: Designed and implemented a complete ML pipeline, including rigorous accuracy evaluation and testing to validate model performance against the MNIST benchmark dataset.
- Core Learning: Solidified understanding of fundamental computer vision and deep learning concepts through practical implementation.

Technical Skills

Programming Languages: Python, Java, C/C++, Haskell
Frameworks & Libraries: TensorFlow, Keras, Google Gemini API, Gradio, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, python-chess
Tools & Platforms: Git, Linux, VS Code, IntelliJ, Jupyter Notebook, Stockfish
Languages: Chinese(native), Portuguese(fluent), English(fluent)